

Swobjects: SPARQLing away at data integration – drinking your data from the source

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Digital Enterprise Research Institute





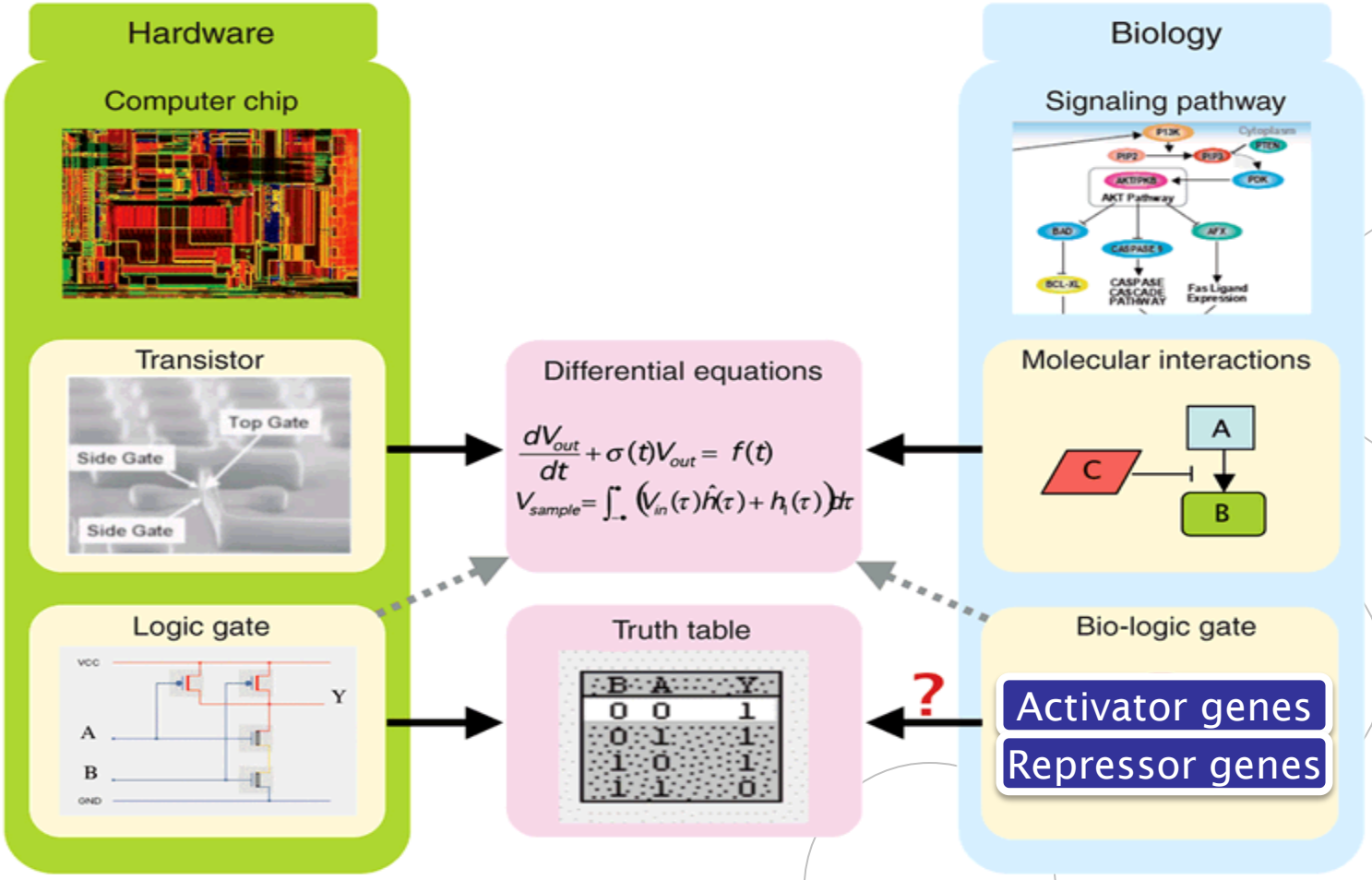
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What is the role for SW in LF?



Biology is becoming an information science!

- Cell and computers are not that different



Src: http://gene-therapy.yolasite.com/process.php

Linked data for Life Sciences



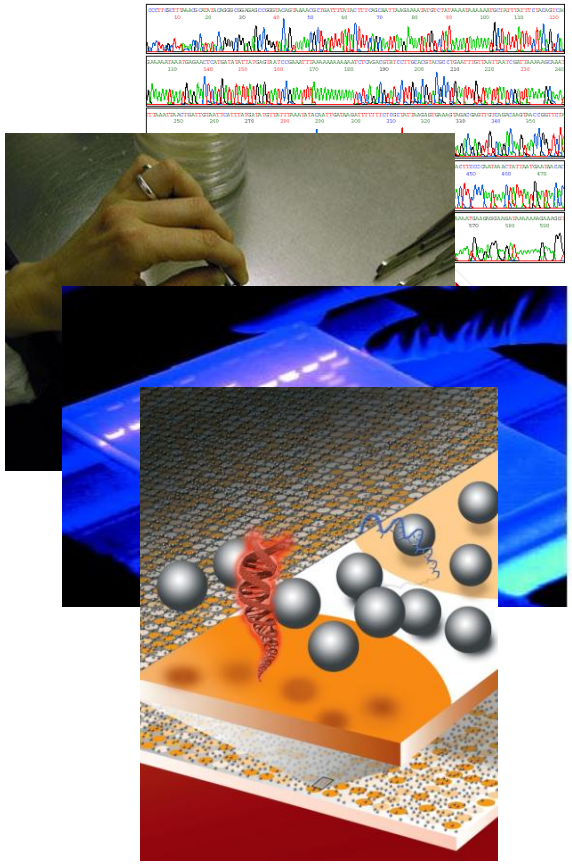




"Knowledge"

Linked data for Life Sciences



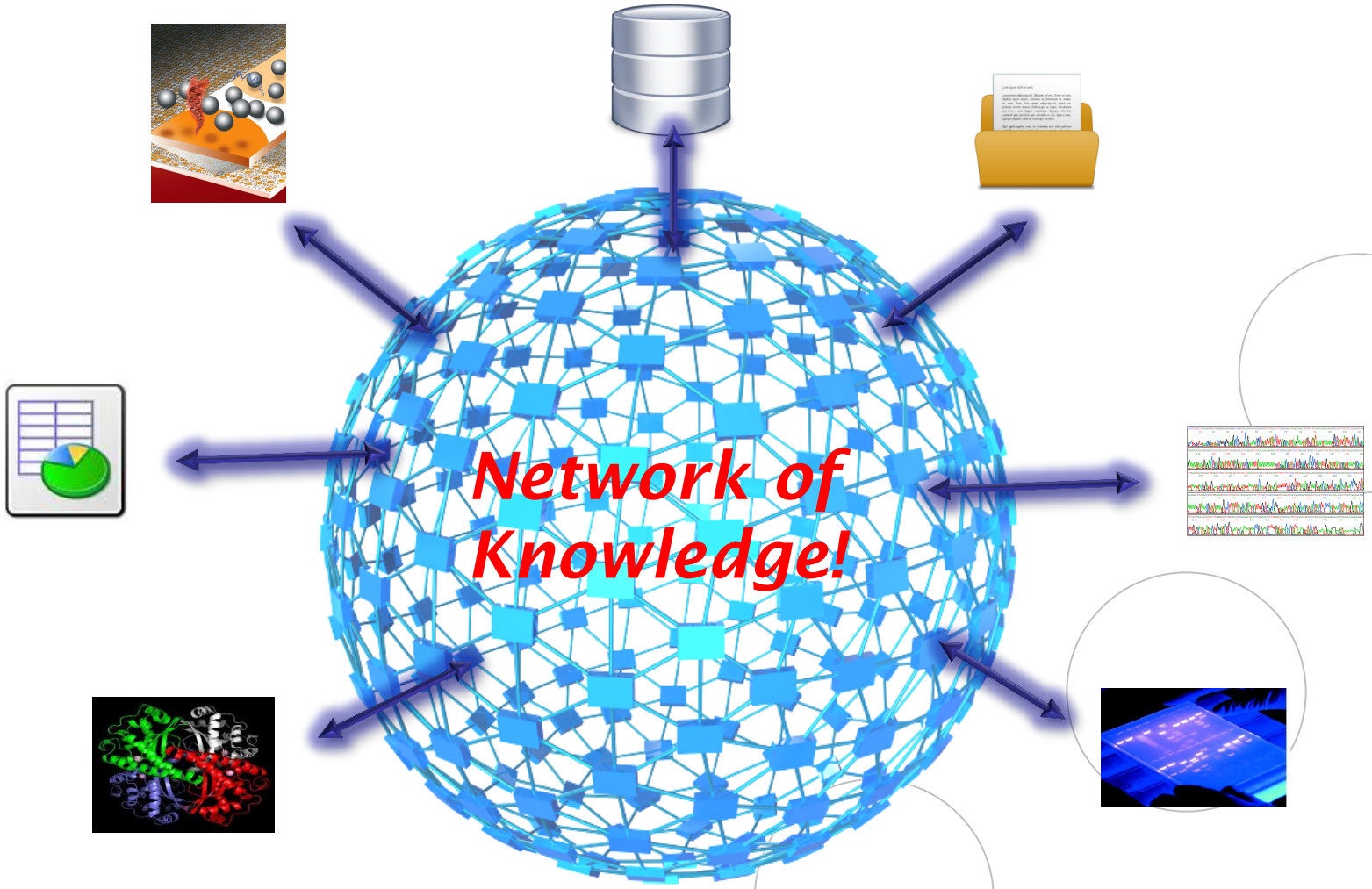




RAW DATA!

Interpretation of Results!

Linked data, Networked Knowledge



Is Linked Data Semantic Web in new clothes?



"Linked Data refers to a set of best practices for publishing and connecting structured data on the Web."
- Chris Bizer, 2009

11 Aug 2011

2009

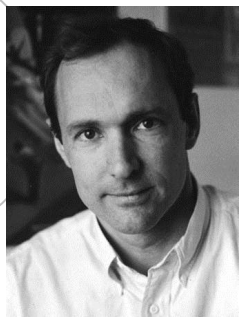
"In the Semantic Web, it is not the Semantic which is new, it is the Web which is new".
- Chis Welty/IBM

2005

We need to "put the Web back in Semantic Web"
- ISWC, Tim Berners-Lee, 2005

2001

Semantic Web is an extension of the current Web in which information is given well-defined meaning …"
- Tim Berners-Lee, 2001



Is Linked Data Semantic Web in new clothes?



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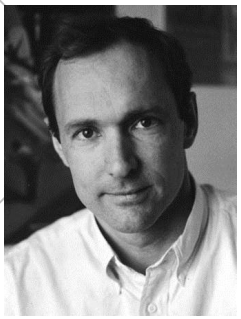
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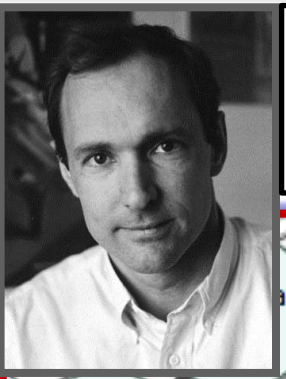


2001

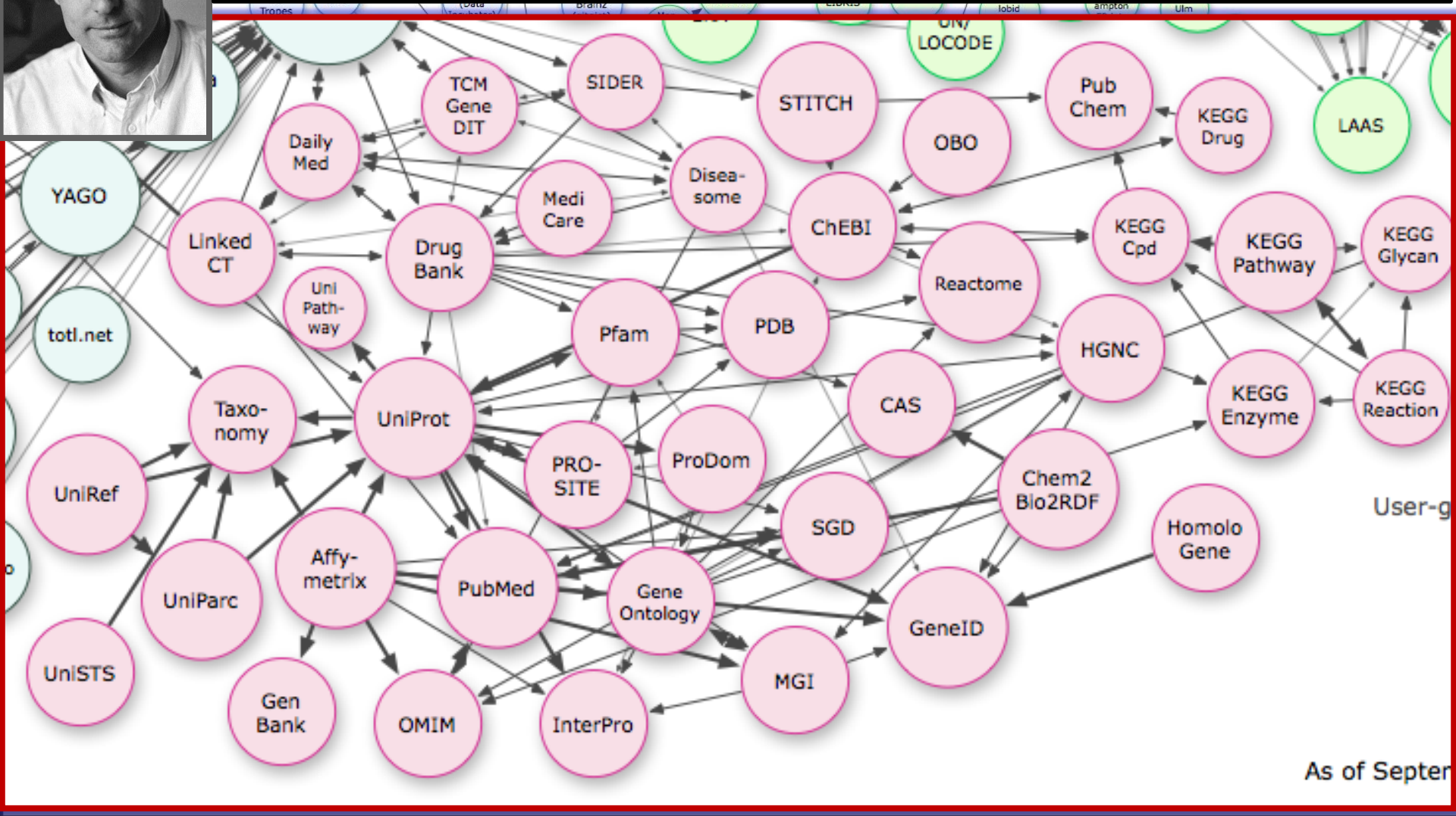
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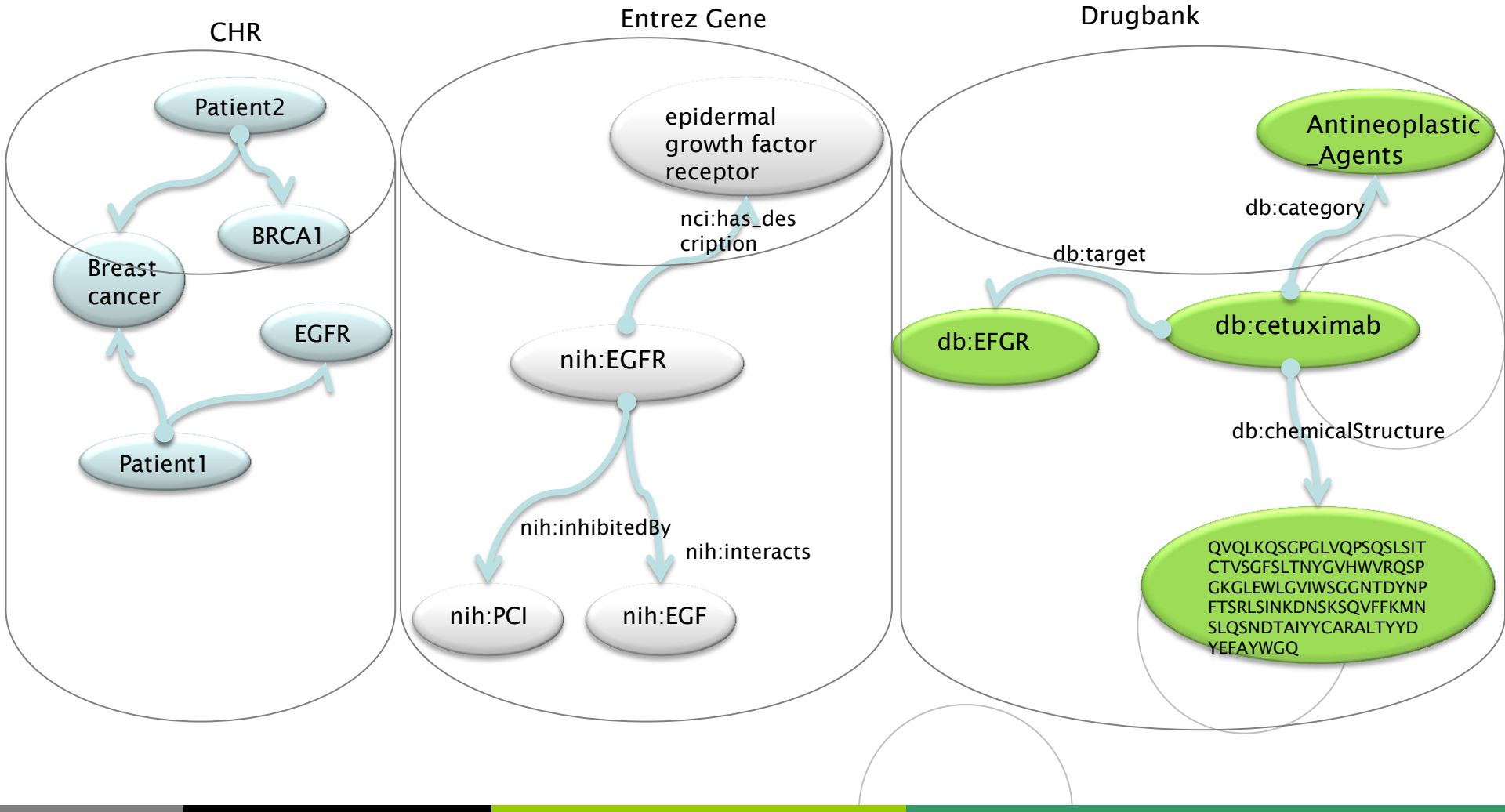
Linked Data – so what?



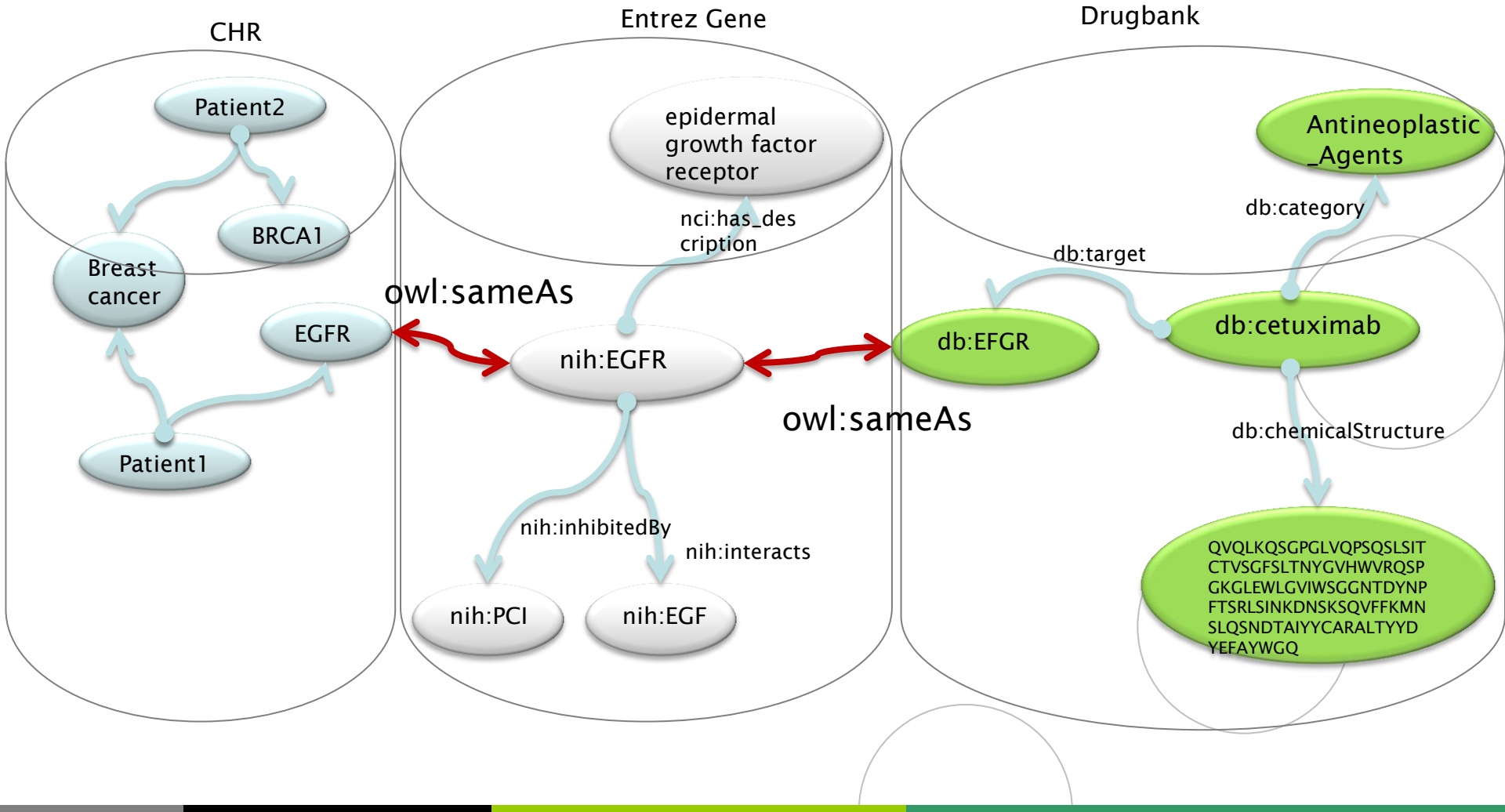
"Life sciences will drive adoption of the Semantic Web, just as high-energy physics drove the early Web."
- BioIT World, 2005



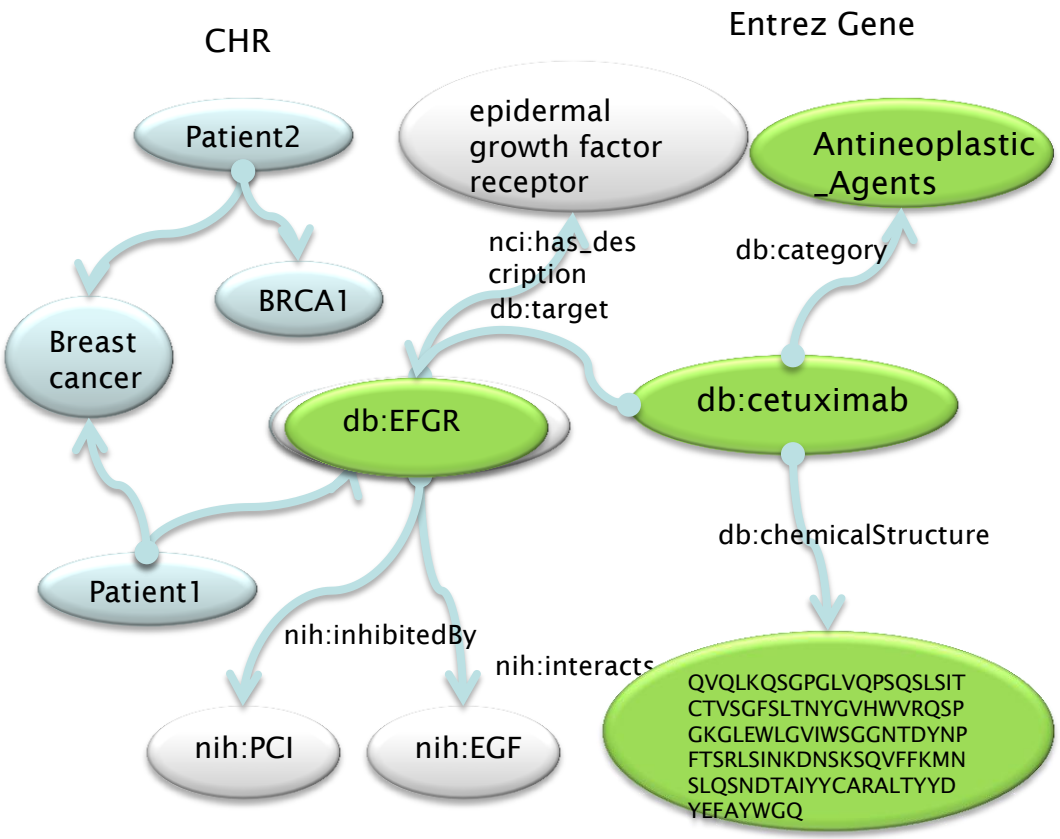
Integrate Data for better drugs



Integrate Data for better drugs

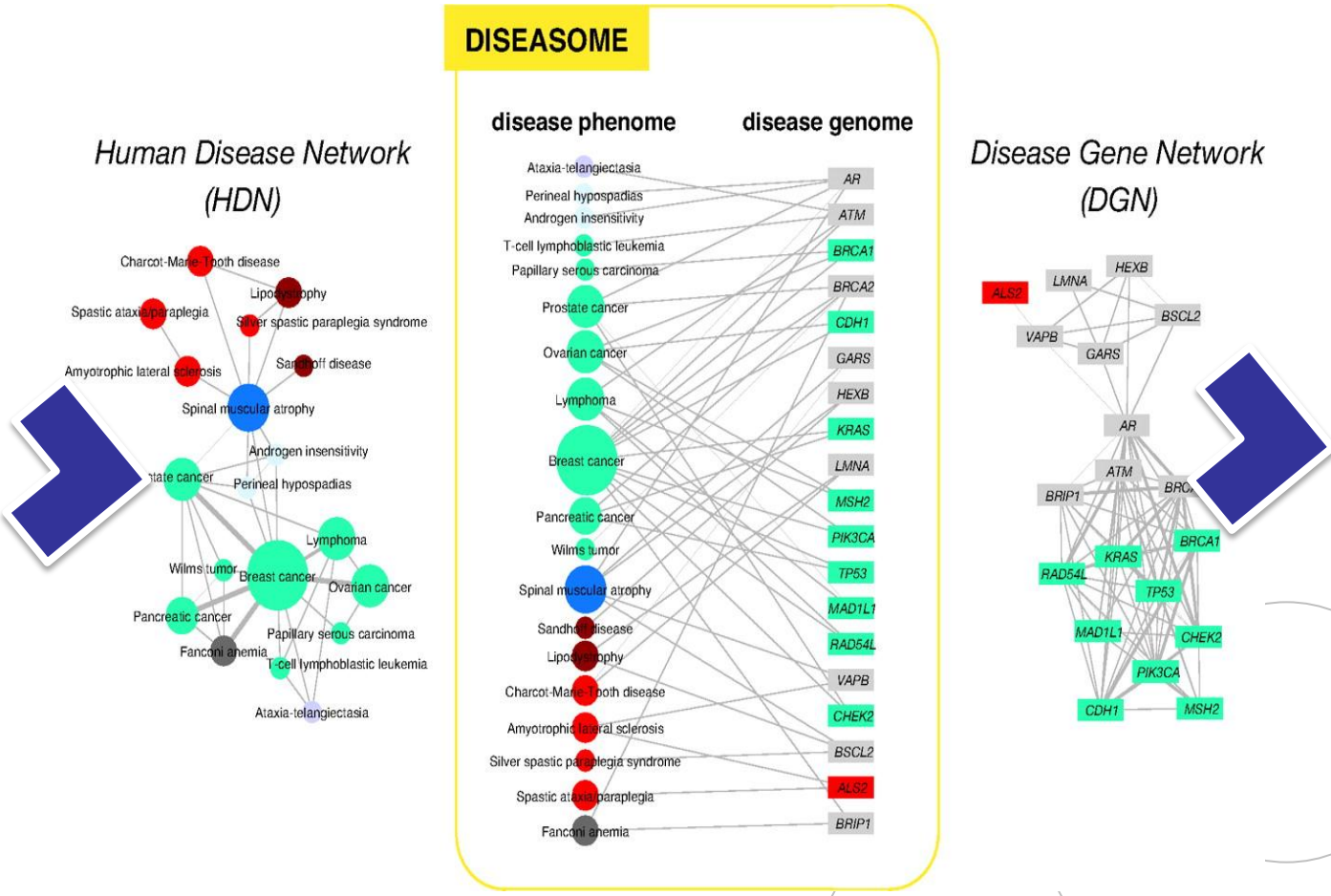


Integrate Data for better drugs



Linked Data – so what?

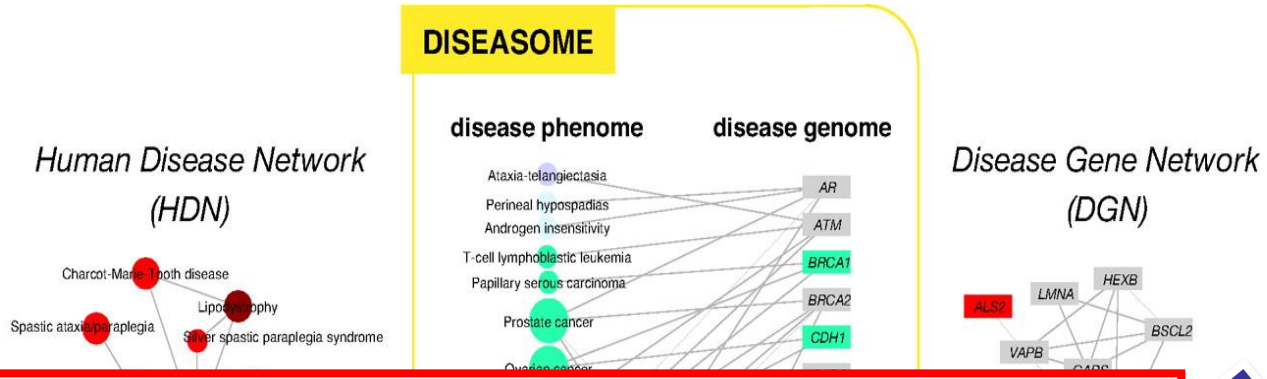
From Clinical Research



Safer and better drugs and treatments

Source: PNAS 104:21, 8685 (2007)

Linked Data – so what?



Safer and better drugs and treatments

Full text access provided to National Univers

nature International weekly journal of science

Home | News & Comment | Research | Careers & Jobs | Current Issue | Archive | Audio & Video | For Auth

Current Issue › Research Highlights › Article

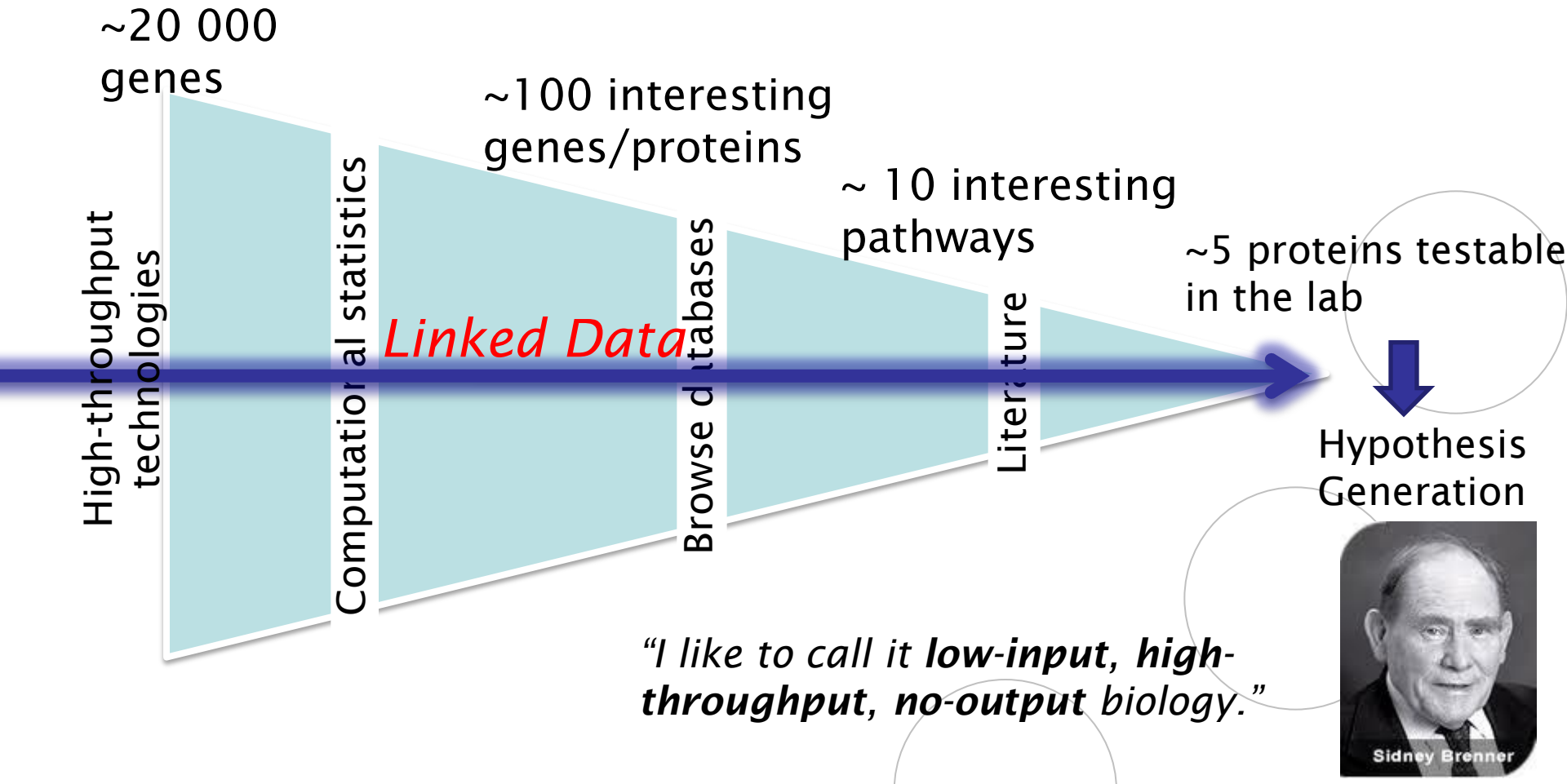
NATURE | RESEARCH HIGHLIGHTS ◂ previous article next article ▸

Cancer biology: Malaria drug shrinks tumours

Nature **472**, 8 (07 April 2011) | doi:10.1038/472008b
Published online 06 April 2011

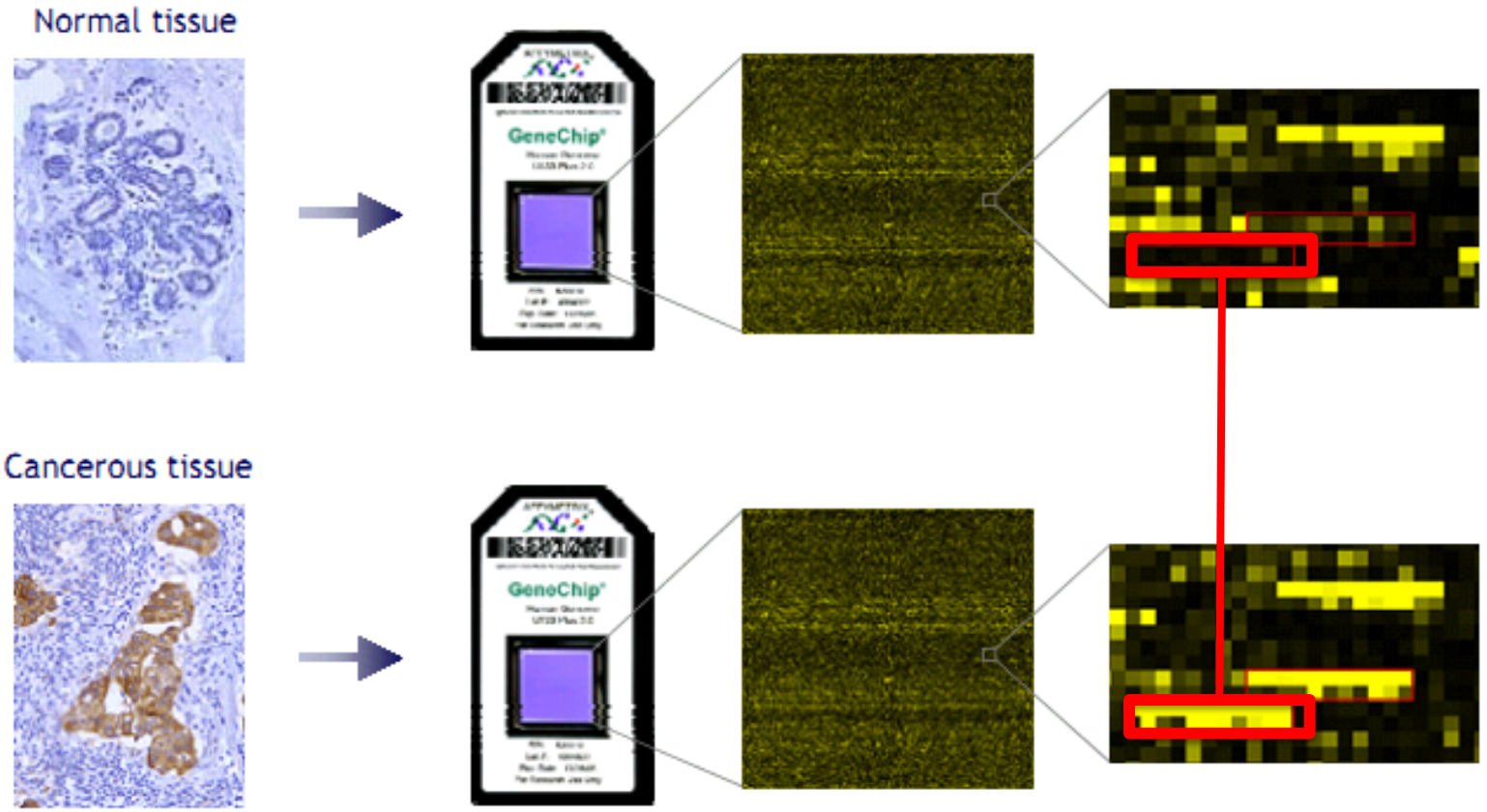
Subject terms: Cancer

The problem with drug discovery

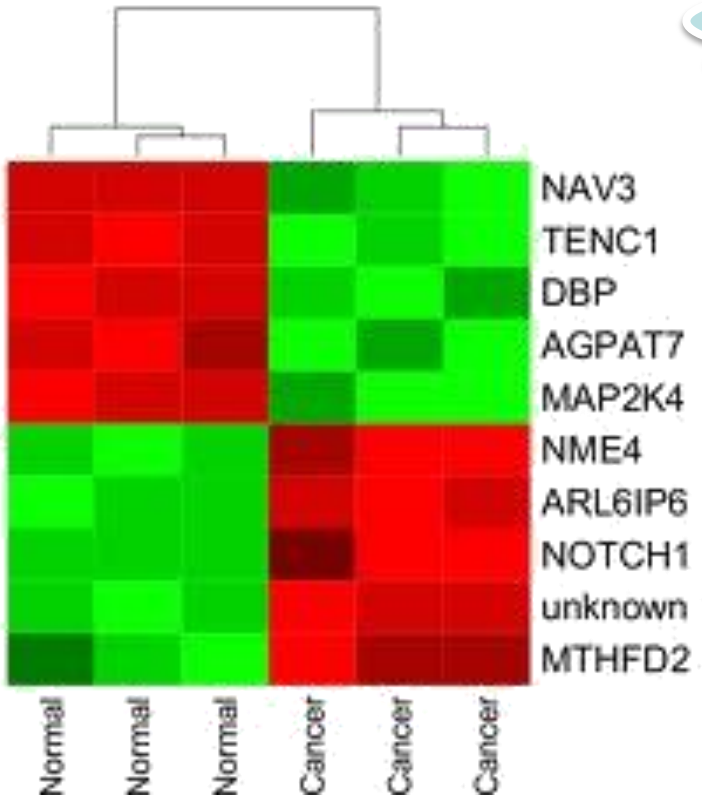


*"I like to call it **low-input, high-throughput, no-output** biology."*

Linked Data for Understanding Cancer



Experimental results + Linked Data = Knowledge



Do these genes share a pathway?

What is the function of each of the cancer genes?

Is there a drug that targets all or most of those genes?

What diseases are genetically Linked to Alzheimer's?

Select samples from Alzheimer's

Select relevant experiments with those samples

Select genes that are over-expressed

Use the genes to query the diseasome

```
PREFIX diseasome: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/
diseasome/> …

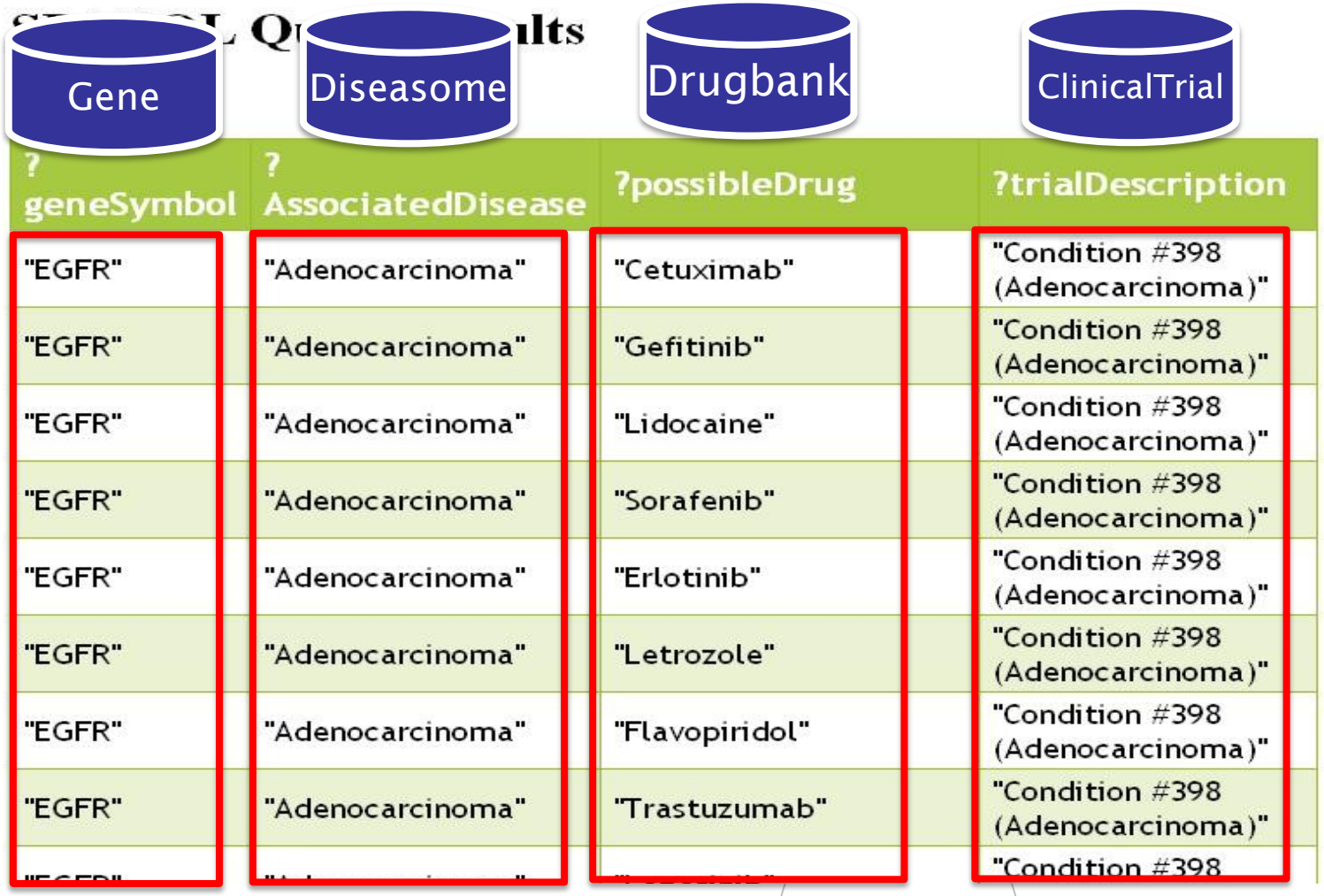
SELECT DISTINCT  ?diseaseName ?geneLabel ?geneName WHERE {
{
        ?sampleList          biordf:patients_have_disease    ?alzheimers .
        FILTER (?alzheimers = doid:DOID_10652 )

        ?experimentSet       dct:isPartOf          ?microarray_experiment
                             biordf:has_input_value          ?sampleList ;
                             biordf:differentially_expressed_gene        ?gene
                             biordf:has_ouput_value ?foldChange .
        ?gene       rdfs:label ?geneLabel .

        ?foldChange          rdf:value   ?foldChangeValue .
        FILTER (xsd:float(?foldChangeValue) > 0)

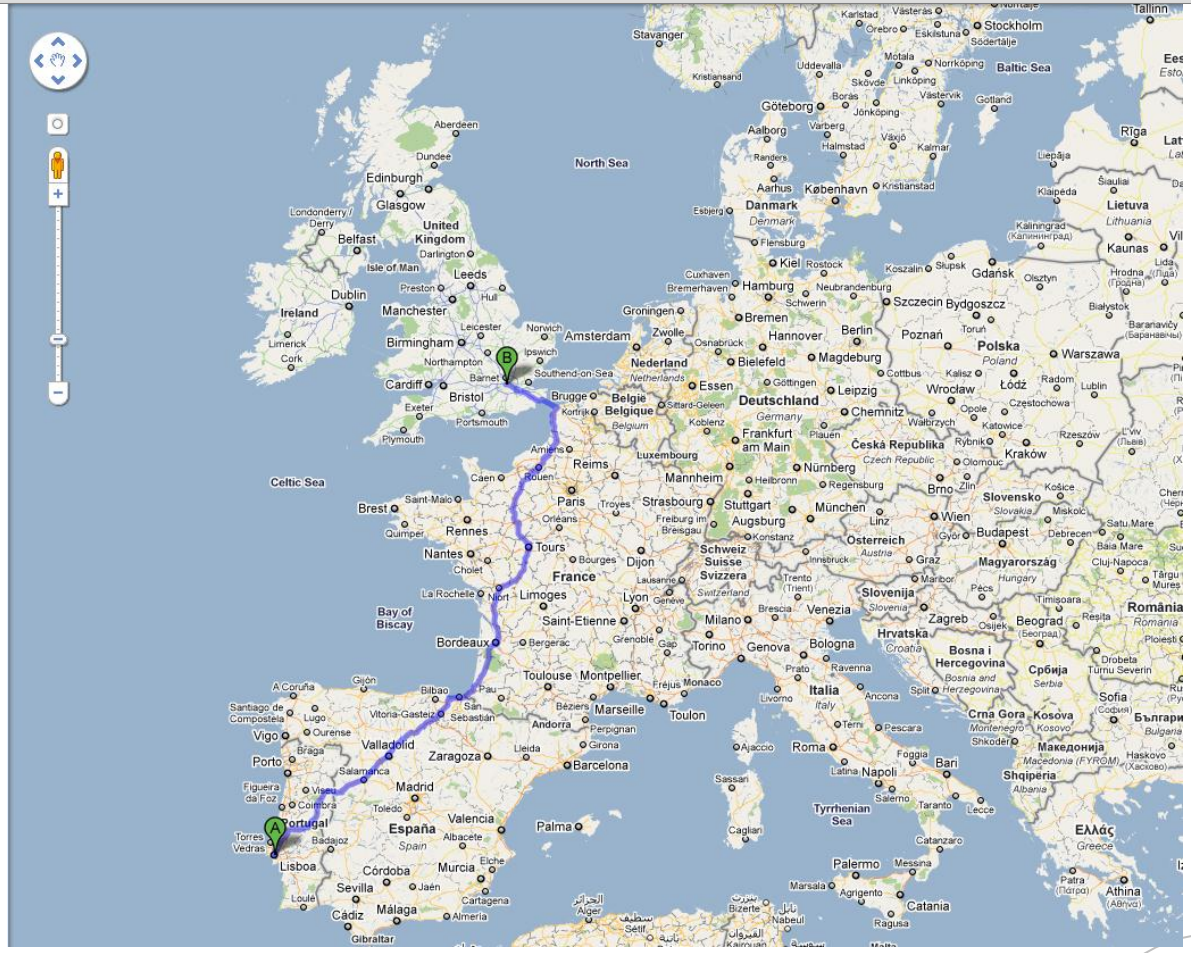

        SERVICE < http://www4.wiwiss.fu-berlin.de/diseasome> {
                ?diseasomeGene rdfs:label ?geneLabel .
                ?disease diseasome:associatedGene ?diseasomeGene.
                ?disease rdfs:label ?diseaseName .
        }
}
```

Some SPARQL federation results



?geneSymbol	?AssociatedDisease	?possibleDrug	?trialDescription
"EGFR"	"Adenocarcinoma"	"Cetuximab"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Gefitinib"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Lidocaine"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Sorafenib"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Erlotinib"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Letrozole"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Flavopiridol"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Trastuzumab"	"Condition #398 (Adenocarcinoma)"
"EGFR"	"Adenocarcinoma"	"Sunitinib"	"Condition #398

An app for Tracking superbugs

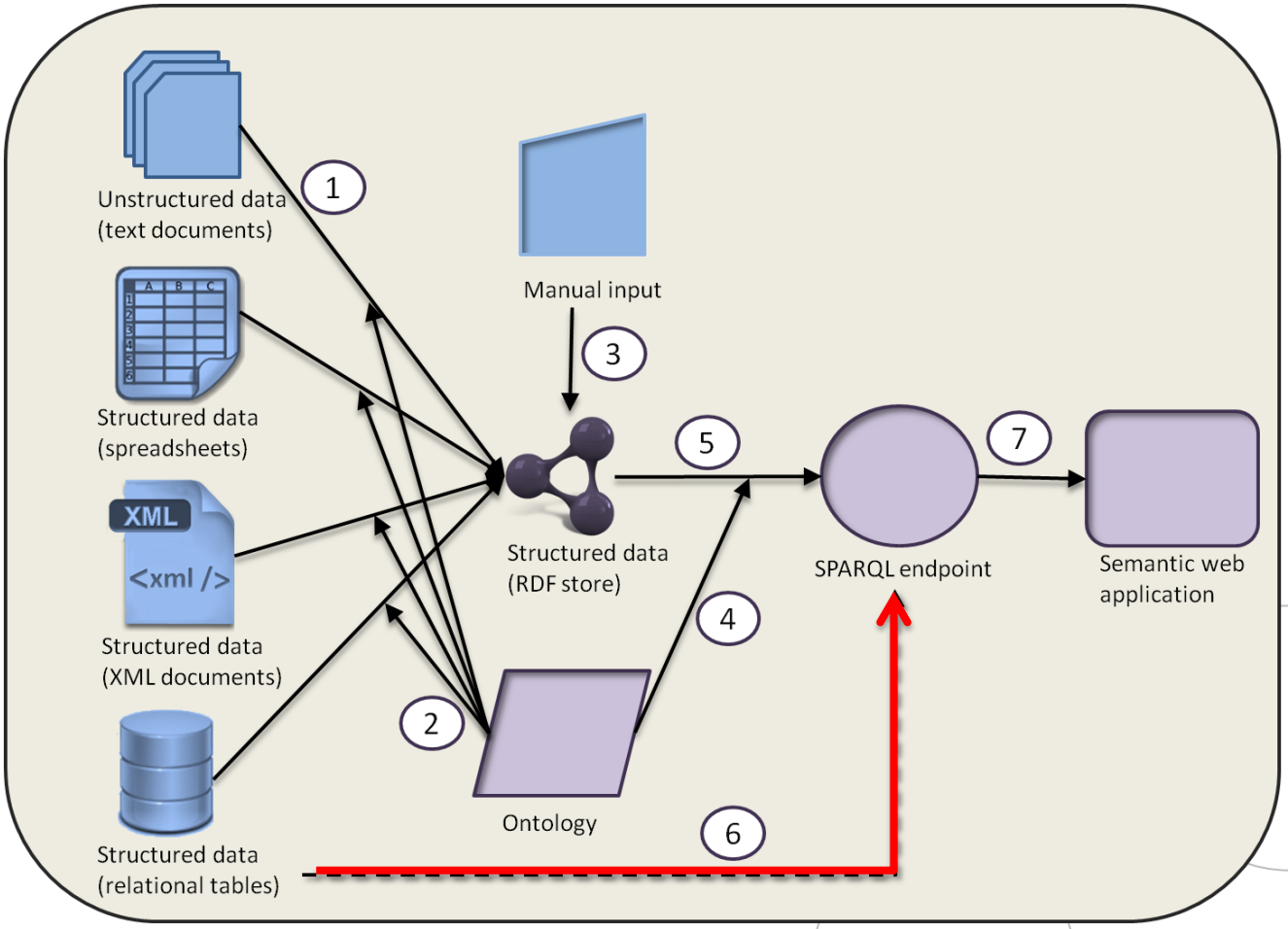


UK superbugs

Portugal superbugs

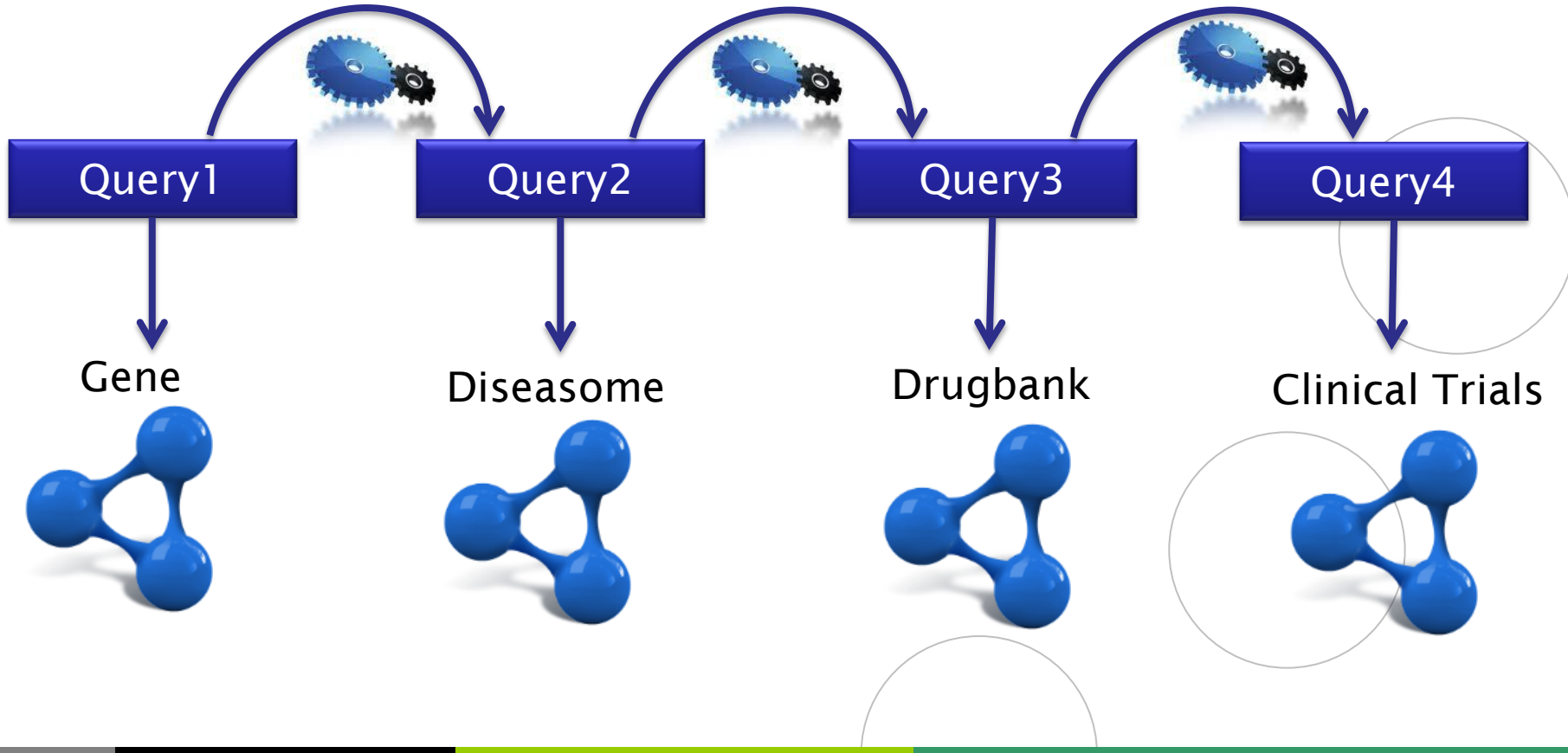
ITOBisolate	arcC	aroE	glp	gmk	pta	tpi	yqiL
h	2	2	2	2	6	3	2
b	2	2	2	2	6	3	2
d	2	4	1	8	4	4	3
j	13	3	1	1	12	11	13
i	6	5	6	2	7	14	5
e	3	3	1	12	4	4	16
f	31	4	1	4	12	1	10
a	31	4	1	4	12	1	10
c	1	65	1	4	12	1	10
g	3	35	48	19	20	26	39

Linked Data Best Practices

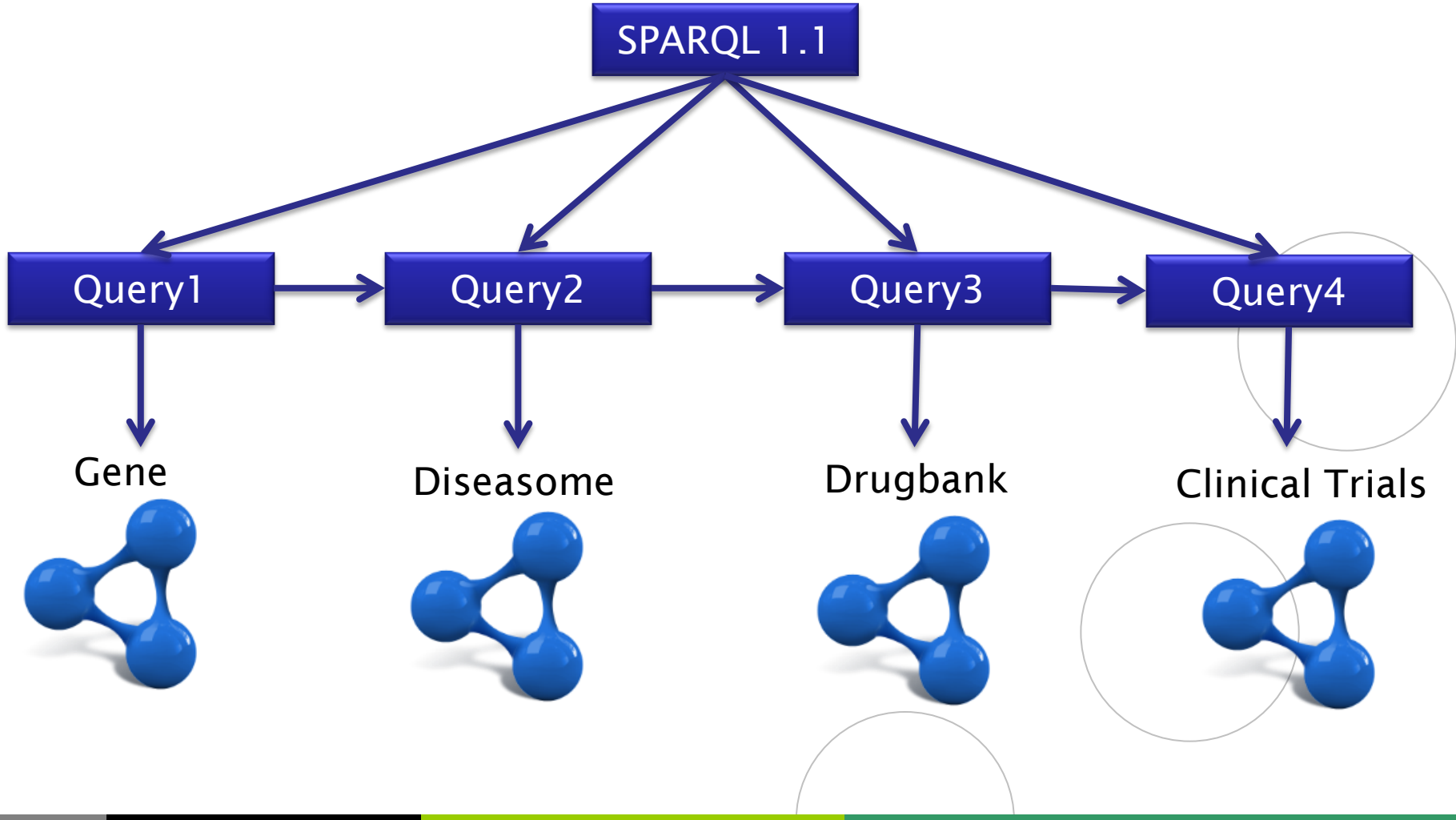


SPARQL as we know it

=> Not very exciting…



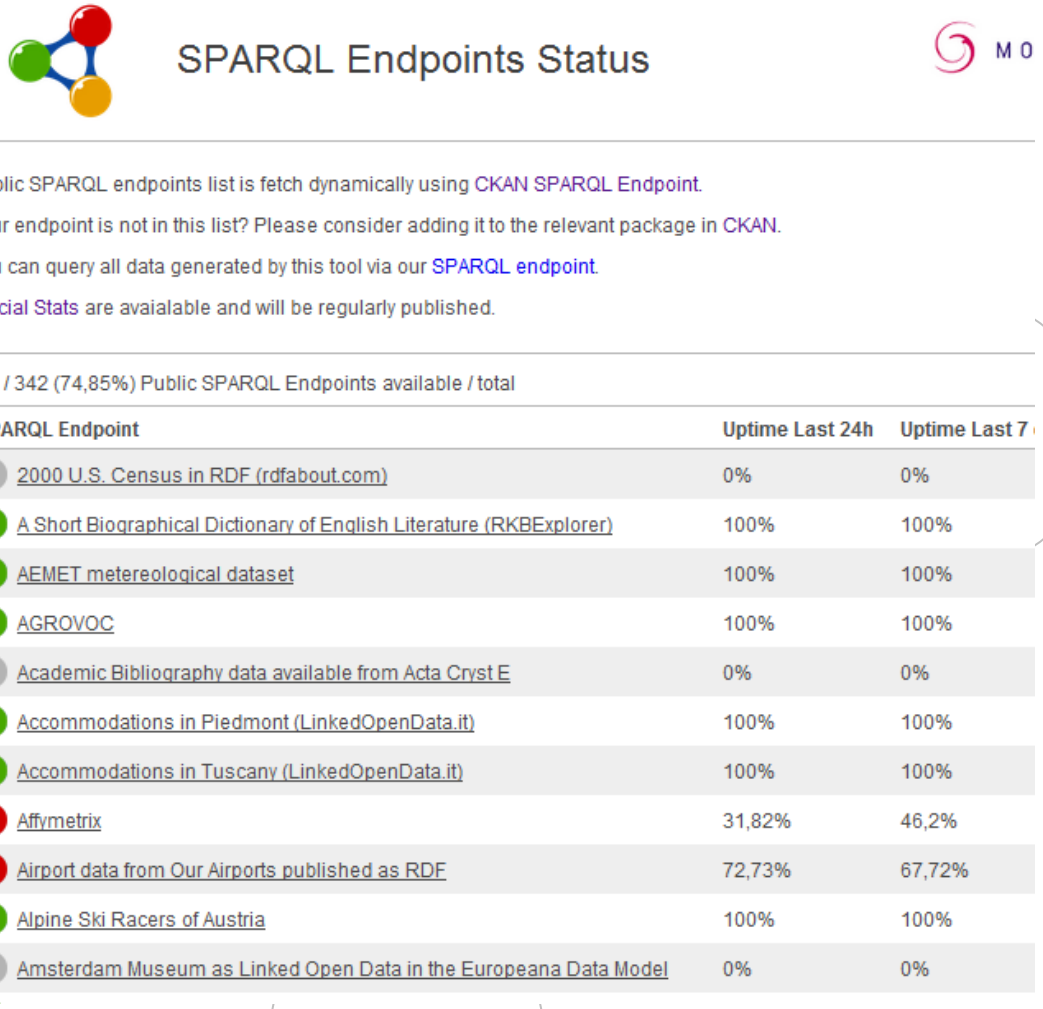
SPARQL as it SHOULD work



Problems in Web of Data land…

- SPARQL endpoint downtime

http://labs.mondeca.com/sparqlEndpointsStatus/



SPARQL Endpoints Status

Public SPARQL endpoints list is fetch dynamically using CKAN SPARQL Endpoint.

Your endpoint is not in this list? Please consider adding it to the relevant package in CKAN.

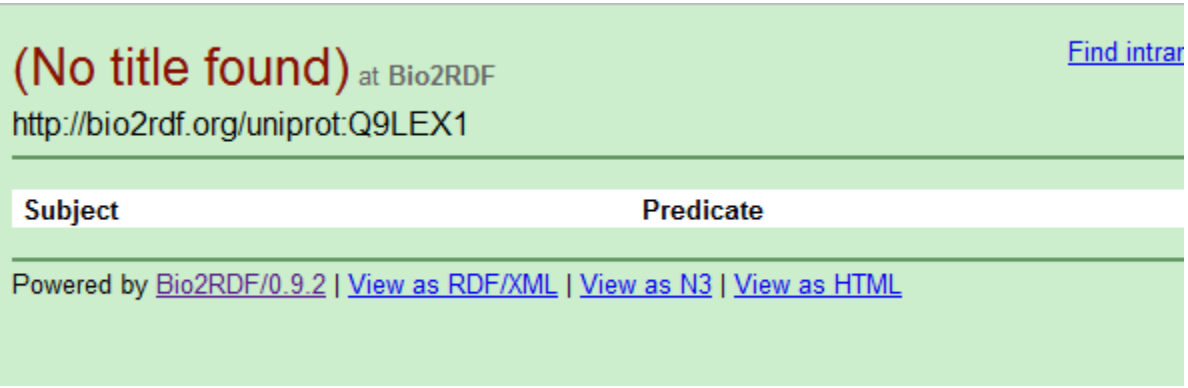
You can query all data generated by this tool via our SPARQL endpoint.

Official Stats are avaialable and will be regularly published.

256 / 342 (74,85%) Public SPARQL Endpoints available / total

SPARQL Endpoint	Uptime Last 24h	Uptime Last 7
2000 U.S. Census in RDF (rdfabout.com)	0%	0%
A Short Biographical Dictionary of English Literature (RKBExplorer)	100%	100%
AEMET metereological dataset	100%	100%
AGROVOC	100%	100%
Academic Bibliography data available from Acta Cryst E	0%	0%
Accommodations in Piedmont (LinkedOpenData.it)	100%	100%
Accommodations in Tuscany (LinkedOpenData.it)	100%	100%
Affymetrix	31,82%	46,2%
Airport data from Our Airports published as RDF	72,73%	67,72%
Alpine Ski Racers of Austria	100%	100%
Amsterdam Museum as Linked Open Data in the Europeana Data Model	0%	0%

Problems in Web of Data land…



- URI that do not dereference

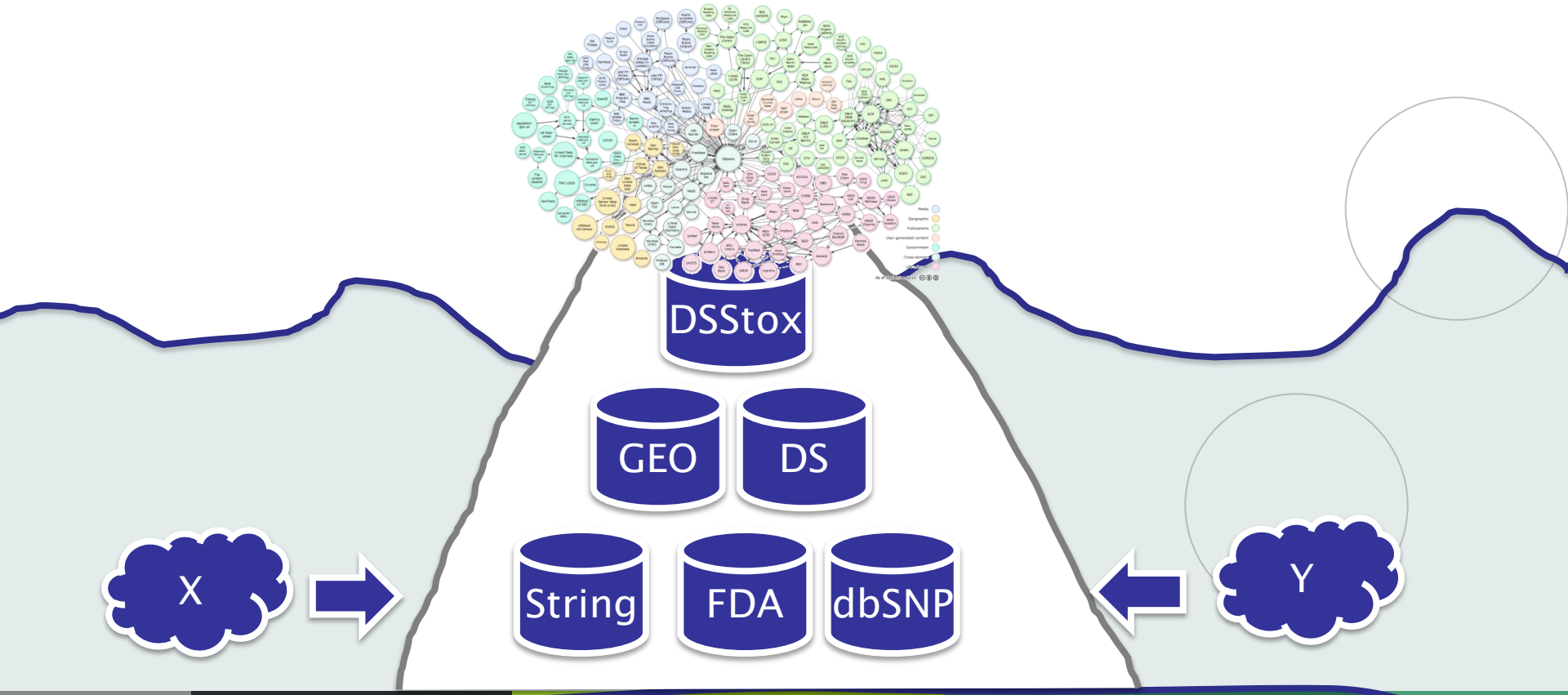
- Orfan URI

Subject	Predicate	Object
http://bio2rdf.org/unigene:At.25576	http://bio2rdf.org/bio2rdf_resource:linkedToFrom	http://bio2rdf.org/affymetrix:11986_at
		http://bio2rdf.org/geneid:825277
	rdf:type?	http://bio2rdf.org/geneid_record:825277
		http://bio2rdf.org/iproclass:Q9LEX1

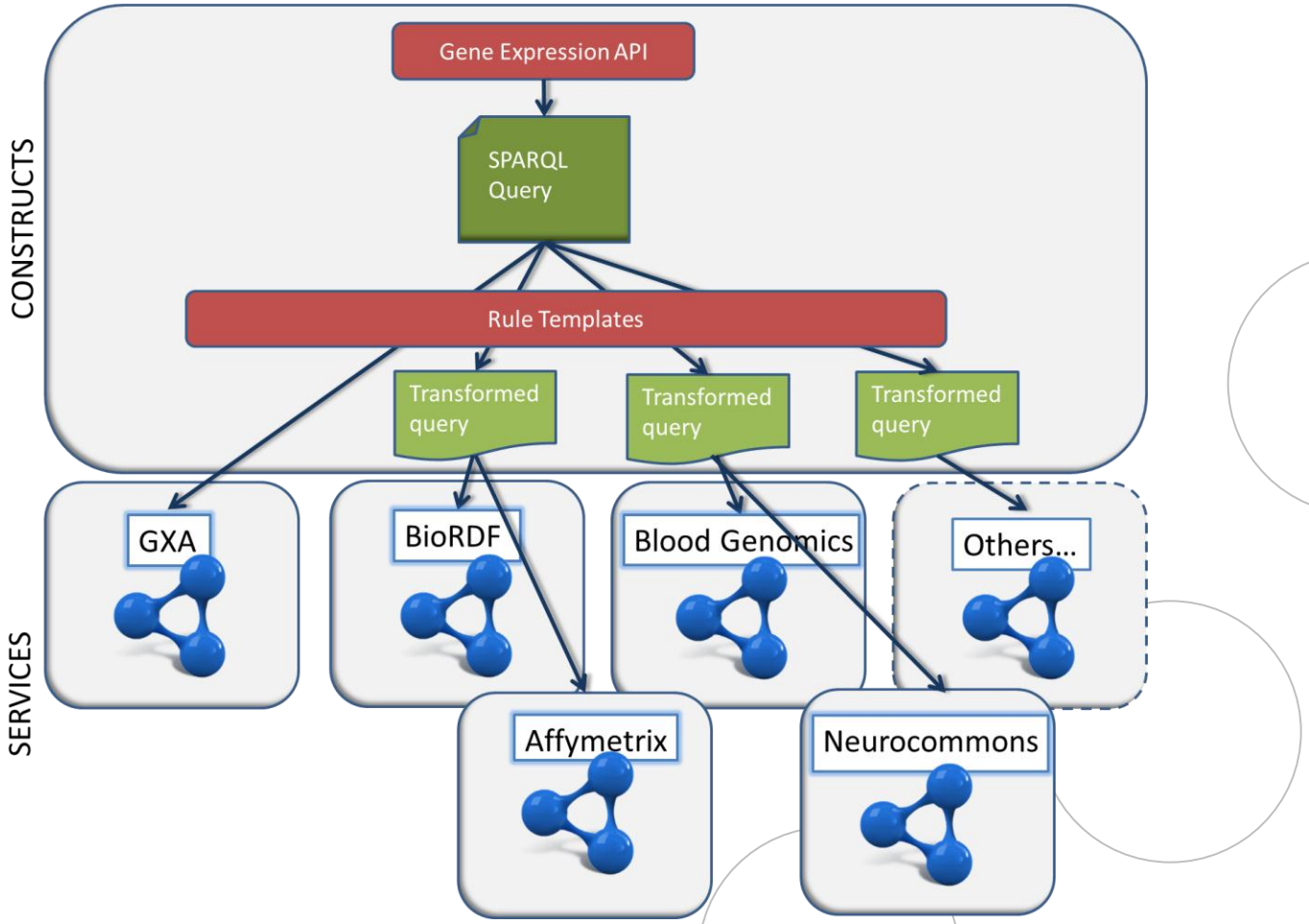
  - No updates… no real incentive to keep the work going beyond POC

Most BioData is Still in Silos

- With Linked Data, we've reached only the tip of the iceberg



How a semantic web API for genomics became possible



SPARQL as SQL

- Converting the query instead of converting the data can create very ugly queries…
- We also want to be good citizens – automated RDB2RDF tools don't give us control over the data so that we can specify ontologies
- It is also not straightforward to link to external URI, for example, bio2rdf

How about SPARQL?

- What if we could use SPARQL to
  - Align the schemas/ontologies
  - Create links between URI
  - Transform values as needed

- And we want to do all this without forcing the data into a triple store, where it would likely become outdated very quickly?

- SWObjects Workshop!